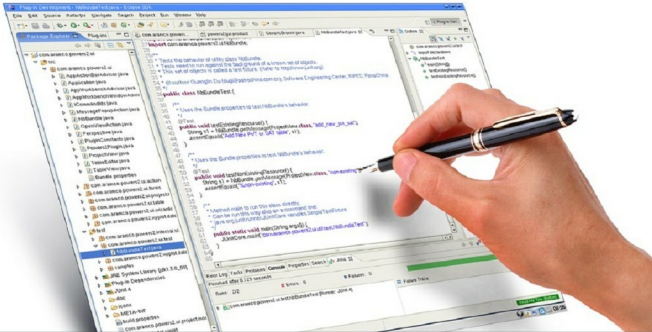


Developing a Web-app Using the Eclipse Editor



This article is the first in a series on developing a Web application called 'Online Banking' using the Eclipse Editor. The application will perform all the normal functions of offline banking. This task will be spread over a number of articles. In this first article, the author covers the basics of what is required to develop the Web-app, starting with Java.

Java is a programming language created by James Gosling from Sun Microsystems in 1991. The first publicly available version of Java (Java 1.0) was released in 1995. The Java virtual machine (JVM) is a software implementation of a computer that executes programs like a real machine.

The JVM is written specifically for a particular operating system, e.g., for Linux. A special implementation is required for Windows.

Characteristics of Java

The aim of the creators of Java was to be able to write a program once and then run it on multiple operating systems.

Java has the following properties.

- **Platform independence:** Java programs use the Java virtual machine as abstraction and do not access the operating system directly. This makes Java programs

highly portable. A Java program (which is standards-compliant and follows certain rules) can run unmodified on all supported platforms, e.g., Windows or Linux.

- **An object-orientated programming language:** Except for the primitive data types, all elements in Java are objects.
- **A strongly-typed programming language:** Java is strongly-typed; the types of variables used must be pre-defined and conversion to other objects is relatively strict. In most cases, it must be done by the programmer.
- **An interpreted and compiled language:** Java source code is transferred into the byte-code format, which does not depend on the target platform. These byte-code instructions are interpreted

by the Java Virtual Machine (JVM). The JVM contains a so called 'hotspot-compiler', which translates performance critical byte-code instructions into native code instructions.

- **Automatic memory management:** Java manages the memory allocation and de-allocation for creating new objects. The program does not have direct access to the memory. The garbage collector automatically deletes objects to which no active pointer exists.

An introduction to the Java editor—Eclipse

Eclipse is an integrated development environment (IDE) for developing applications using programming languages like Java and others such as C/C++, Python, Ruby, etc.

The Eclipse platform provides the